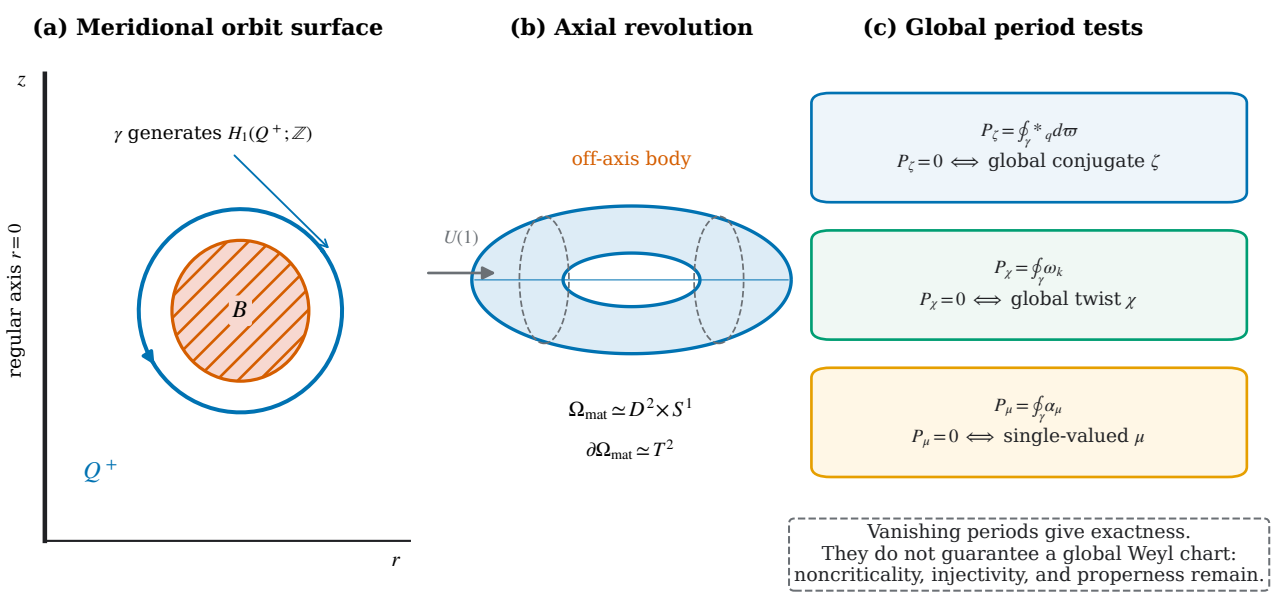
(a) Meridional orbit surface
(b) Axial revolution
(c) Global period tests
z
regular axis r=0
γ generates H1(Q+;ℤ)
B
Q+
r
off-axis body
U(1)
Ωmat ≃ D2×S1
∂Ωmat ≃ T2
Pζ = ∮γ *g dϖ
Pζ = 0 ⟺ global conjugate ζ
Pχ = ∮γ ωk
Pχ = 0 ⟺ global twist χ
Pμ = ∮γ αμ
Pμ = 0 ⟺ single-valued μ
Vanishing periods give exactness.
They do not guarantee a global Weyl chart:
noncriticality, injectivity, and properness remain.